



MANIPAL ACADEMY OF HIGHER EDUCATION

Mid Sem DAA exam -2025

DESIGN AND ANALYSIS OF ALGORITHMS [CSE 2222]

Marks: 30

MCQ

Answer all the questions.

1) What is the pseudocode to find the minimum element in the given array?

$maxval \leftarrow A[0]$	$maxval \leftarrow A[0]$	$minval \leftarrow A[0]$
for $i \leftarrow 1$ to $n - 1$ do	for $i \leftarrow 1$ to $n - 1$ do	for $i \leftarrow 1$ to $n - 1$ do
if $A[i] < maxval$	if $A[i] > maxval$	if $A[i] > minval$
$maxval \leftarrow A[i]$	$maxval \leftarrow A[i]$	$minval \leftarrow A[i]$
return $maxval$	return $maxval$	return $minval$

2) Consider the following pseudocode for the sequential search algorithm.

```

i ← 0
while i < n and A[i] ≠ K do
    Missing statement
if i < n return i
else return -1
    
```

What is the missing statement in the above code snippet?

[if A\[i\] == K i=i+1 i=i-1 No statement is required.](#)

3) _____ is the solution of following recurrence relation.

$T(n) = 2T(n-1) + 1$ for $n > 0$,

$T(0) = 0$

[2ⁿ-1](#) [2n](#) [n](#) [2\(n-1\)](#)

4) What's the best-case time complexity of Bubble Sort algorithm with early termination?

[O](#) [O\(log n\)](#) [O\(n\)](#) [O\(n log n\)](#)

5) Raj wants to do a round trip (ignoring the directions) between the cities Manipal, Udupi, Shimoga, Bangalore, Mangalore, and Davangere, starting from Manipal and ending at Manipal. He wants to find the shortest possible round trip. This is a Salesman Problem (TSP) using Brute Force approach to make sure that he doesn't pass through any of the roads multiple times. Provided these conditions:

[6](#) [120](#) [720](#) [36](#)

6) When Brute string matching algorithm is used on a pattern of length 10 and text of length 100, what would be best case and worst case number of character comparisons?

[100, 1000](#) [10, 910](#) [50, 500](#) [50, 910](#)

7) List all the keys of the given array that will require the largest number of key comparisons when searched for by using the binary search, {3, 14, 27, 37, 42, 55, 67, 78, 85, 91, 95, 100}

[3 and 85](#) [3 and 42](#) [55 and 85](#) [42 and 55](#)

- 8) What is the time efficiency of

$$T(n) = 8T\left(\frac{n}{2}\right) + 1000n^2$$
 after applying the Master's method.

$\Theta(n)$ $\Theta(n^3)$ $\Theta(n^2)$ $\Theta(n^4)$

- 9) Strassen's algorithm requires how many scalar multiplications to multiply two 8×8 matrices.

343 512 256 216

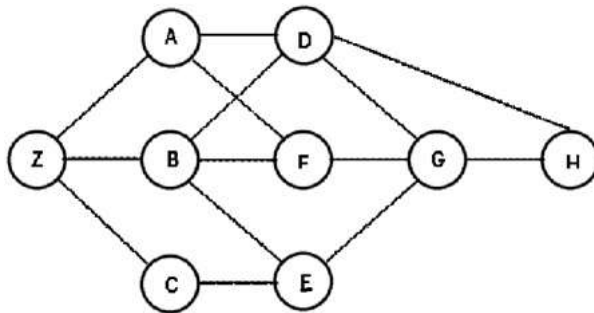
- 10) Large integer multiplication using divide & conquer method for multiplying two 16-digit numbers requires how many single-digit multiplications. (0.5)

256 81 64 27

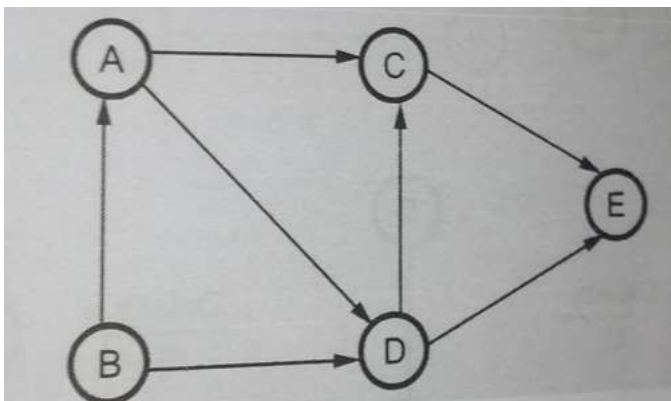
DESCRIPTIVE

Answer all the questions.

- 11) The key aspect of algorithm analysis involves the identification of the most important operation of the algorithm, called the basic operation which core mechanism of dot(.) product is adopted to find the product $C=A*B$ of two $N \times N$ matrices, A and B. Write an algorithm to find the multiplication of two of times the algorithm's basic operation is performed.
- 12) Sort the characters of the string "SELECT" in ascending order using Selection Sort algorithm by clearly showing all the steps. Compute the number of comparisons required to obtain the sorted array.
- 13) Traverse the graph shown in figure below by breadth first search. Write the traversal order and construct the corresponding breadth first search tree. Show the order of nodes in alphabetical order. Also list the cross edges encountered.



- 14) How to find the topological sorting of a given Directed Acyclic Graph(DAG) using source removal method? Apply this method to find the topological sorting of the given DAG.



- 15) Apply quicksort on the list 9, 7, 5, 11, 12, 2, 14, 3, 10, 6 to sort in ascending order using the Hoare partition by considering the first element as a pivot.
- 16) Discuss AVL trees along with its properties and drawbacks. Construct AVL tree by showing the successive insertion of key elements and rotations involved. The key elements are 15, 17, 19, 25, 29, 32, 37, 40.
- 17) Sort the list 82, 90, 10, 12, 15, 77, 55, 23 in ascending order using Heap Sort. Show all trees in detail during the bottom-up heap construction and sort.